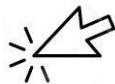


github.com/sailuh/kaiaulu



Refactoring Kaiaulu Architecture

By: Cohen Ruport, Colby Fuke, Gi Young (Michelle)
Back and John Flinn

What is Kaiaulu?

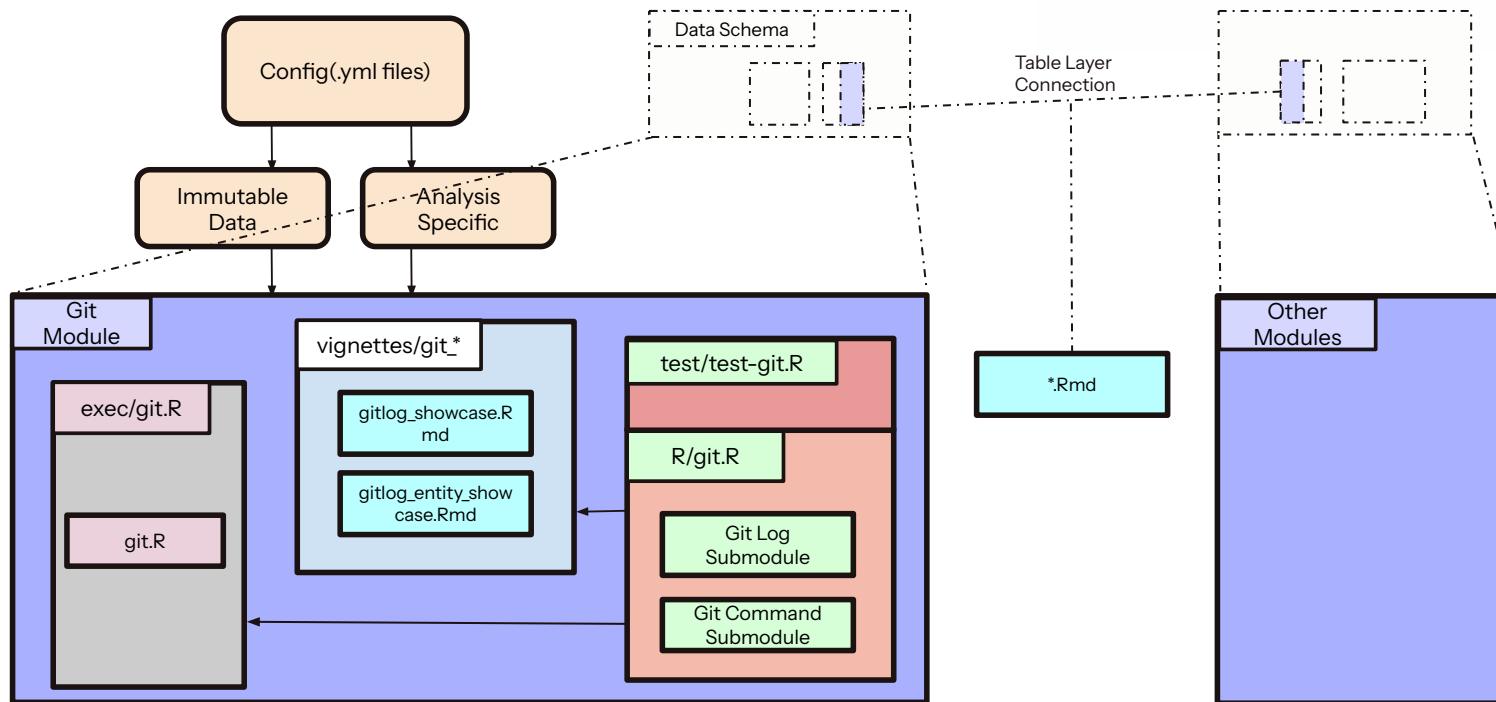


Kaiaulu is an R package and common interface that helps developers in mining software repositories to better understand the evolving software development communities and the artifacts they produce.

Problem Statement:

With many short term contributors (e.g. ICS students at UHM), Kaiaulu has grown with many added features. This, in turn led to Kaiaulu accruing technical debt and inconsistent internal interfaces, motivating the need for architectural refactoring.

Technical Solution Overview



Major Accomplishments

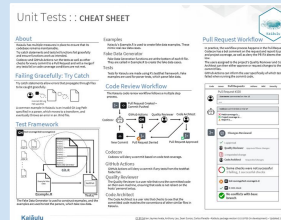
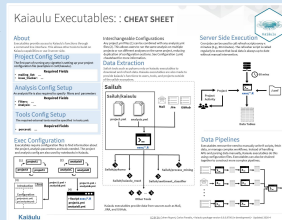
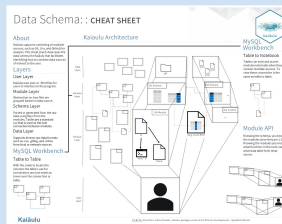
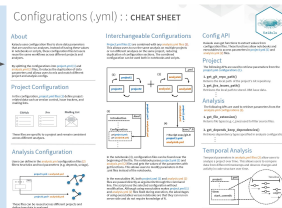


Project Refactoring

- Config Files
 - Split configuration .yaml files into project and analysis
- Data Schema
 - System for documenting tables and their merges
- Executable Files
 - Created new executables and standardized existing ones
- Unit Tests
 - Created new tests, and implemented new types of tests

Cheatsheets

- We each created our own cheatsheets and saved them to Kaiāulu's cheatsheet repo
- This helps users and developers understand how Kaiāulu works



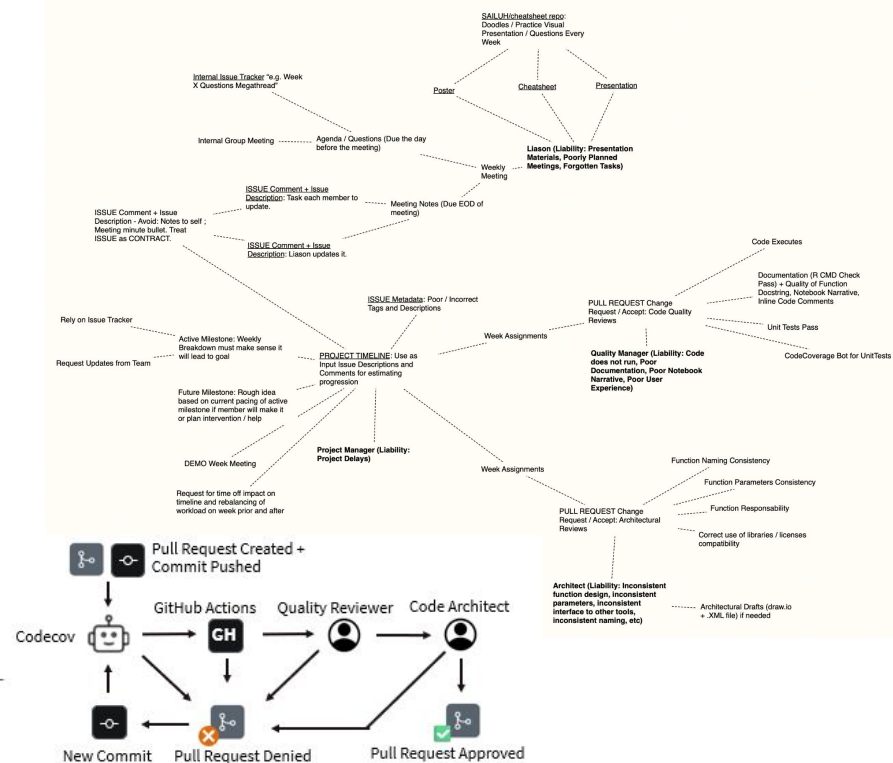
Quality Assurance Metrics



Roles

- Code Architect: Design and style consistency in the codebase
- Quality Manager: Runs code with unit tests
- Project Manager: Timeline management
- Liaison: Communication with the sponsor and group members

Roles can be transferred to other members of the group depending on their background and the specific section of the code



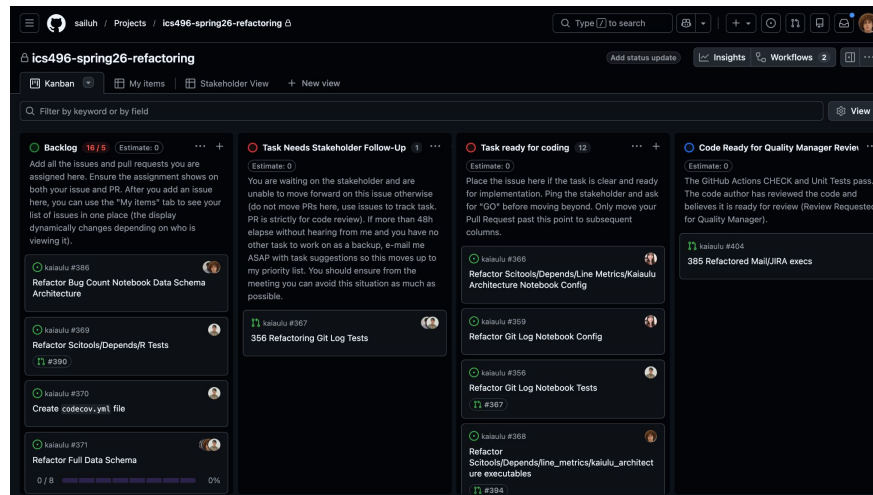
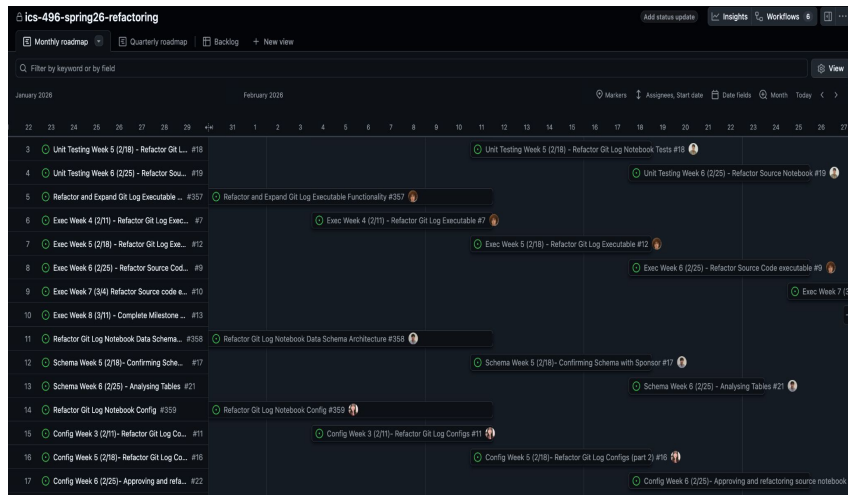


Calendar Milestone Tracker

- Overcomplicated milestones by having separate issues to track progress

Kanban Issue Tracker

- Linked directly to issues
- Easy to track progress



Reflections



Challenges

- Communication
 - Time zones
 - Vague project specifications
 - Team mates
- Effective teamwork
 - Project work is disconnected from each other
- Timeline management
 - Falling behind on deadlines
 - without a end product

Key Takeaways

- Meeting preparedness
 - Detailed note taking
 - Specific questions with context
- Effective teamwork
 - Moving through modules as a group
- Timeline management
 - A collaborative effort to support one another and stay on track with project timelines

Next Steps



Configs

- The configurations have been split into project and analysis .yaml files so the next step is to fully integrate this new system across all the remaining Kaiaulu notebooks.



Data Schema

- Expanding the schema to additional modules in Kaiaulu, using the documentation system laid out in the main branch.



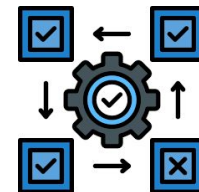
Exec

- Continue expanding Kaiaulu's data pipelines through creating more exec files



Unit Testing

- Expanding the tests into more .R files to test their functionality, and implement other code stability features into Kaiaulu



Acknowledgements



Thank you to our sponsors Carlos and Rick for their guidance, feedback, and support throughout this project! 😊



Carlos Paradis

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Graduate of UHM



Rick Kazman

Professor in
Shidler School of
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THANK YOU!

